Fuzzy Sets opn writeup

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Date

Title: implement Union - inter

Hardware Requirement: Computer, linux (Ubuntu).

Software Requirement: Jupyter Notebook, pycharm IDE.

Theory:

Fuzzy logic: Any particular event which do not represent any of the exact value is fuzzy called fuzzy logic.

Fuzzy Sets: Collecⁿ of element where each element has a membership ranging betⁿ 0 & 1.

if A is fuzzy set, then its membership function is represented as,

$$\mu A(x) \in [0,1]$$

where 0 - no membership, 1 - full membership

0<x<1 → partial membership.
val betn

Fuzzy logic working:

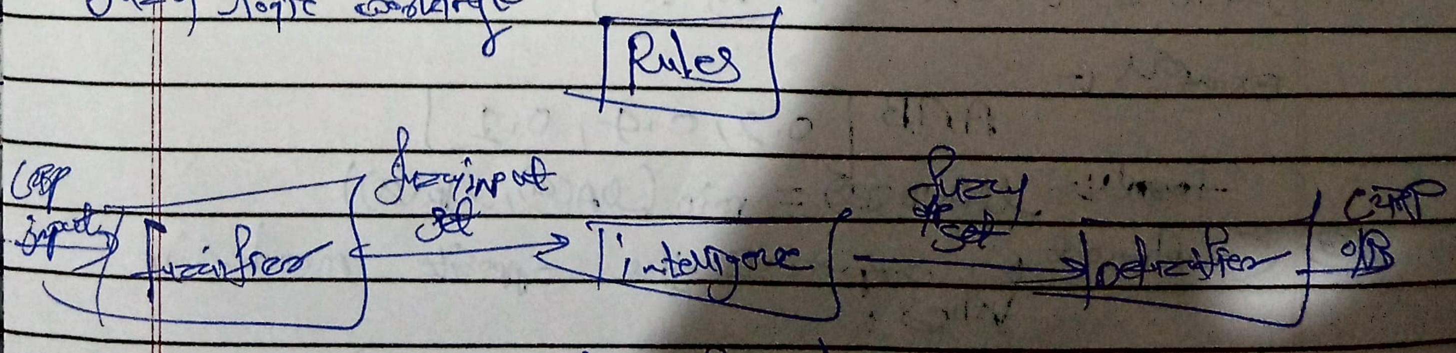

fig - fuzzy logic

Date

Fuzzification: turns exact numbers into fuzzy category

Rulebase: A set of 'If-then' rules logic provided
(e.g. "if its warm, then sets fan to medium"

inference engine: matches the current situation against the rules
see, which ones apply

Defuzzification: Turns those fuzzy results back into one
precise action

O/p results:

① Union: The union of two fuzzy sets A & B obtained by
taking the max membership value of corresponding elements

$$\mu_{A \cup B}(x) = \max(\mu_A(x), \mu_B(x))$$

Example:

if: $A = [0.2, 0.4, 0.6]$
$B = [0.3, 0.5, 0.1]$

then

$A \cup B = [0.3, 0.5, 0.6]$

② Intersection: obtained by taking minimum membership value

example:

$A \cap B \; [0.2, 0.4, 0.1]$

formula: $\mu_{A \cap B}(x) = \min(\mu_A(x), \mu_B(x))$

③ Complement: represents the opposite membership
values.

formula: $\mu_{A^c}(x) = 1 - \mu_A(x)$

Example:-

If: $A = [0.2, 0.4, 0.6]$

then

$A' = [0.8, 0.6, 0.4]$

Application:-
- Artificial intelligence
- decision support systems
- medical diagnosis
- pattern Recognition
- robotics & Automations

(4) diffⁿ:-

formula:-

$A - B = A \cap B'$

~~Human~~ formula:-

$\mu_{A-B}(x) = \min(\mu_A(x), 1 - \mu_B(x))$.

This operation gives membership values that belong to set A. but not to set B.

fuzzy relⁿ:- formed by Cartesian product of two fuzzy sets, it represents the relⁿ betⁿ elemⁿ of two fuzzy sets

Cartesian product:- Cartesian product of fuzzy sets A & B is ~~defined~~

$R(x,y) = \min(A(x), B(y))$

where:-
- $R(x,y)$ - membership val. of Relⁿ R.
- minimum operator is used to generate fuzzy relⁿ matrix

max-min Composition:- used to Combine two fuzzy relations

if R & S are fuzzy sets relⁿ their Composition:-

$T(i,k) \max_j \min(R(i,j), S(j,k))$

Steps:-
- Take min valᵘ betⁿ Corresponding elements
- find max among these min values.
- Store results in the Composed relⁿ matrix
- used in fuzzy inference engine systems & decision-making applications